Data Mining in Education: Discussing Knowledge Discovery in Database (KDD) with Cluster Associative Study

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ABSTRACT

Data Mining, being an emerging field has ties with ever-increasing knowledge base, contributing to historical data trends and patterns as well as providing evidence to predict future behaviors. With the adoption of data mining techniques and their thorough implications, the knowledge base could be enhanced and a futuristic strategic management tool could be formulated in the form of modern data mining techniques. The technique and terminology used in the paper involves extracting knowledge from large datasets. In this paper, cluster sampling associated with data mining is used to dig out a beneficial way of tackling data mining steps. With these combined data mining efforts, the aim of highlighting the significance of data mining could be amalgamated with data warehouses and the steps that need to involve selecting, transforming, mining, and output evaluation to get beneficial or desired results. Furthermore, a mild emphasize is put on the cluster sampling data mining technique, steps, and its implication to enhance knowledge, explore new terms, and broaden our output scenarios so that there is a better understanding of how to move along the data mining channel. The objective of this study is to explain the basics of data mining and provide a way to further this research over data mining with cluster sampling and Knowledge discovery is data bases (KDD). For the said purpose, two clusters of the population of China are used in this study. One of which includes the percentage of students enrolled in the primary schools; and the other includes percentage of students enrolled in secondary schools in China. The steps of cluster sampling technique are followed to give a better understanding of the data mining approach.

CCS CONCEPTS

• **Computing methodologies** → Modeling and simulation; Model development and analysis.

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KEYWORDS

Educational data mining, cluster associative study, artificial intelligence

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1 INTRODUCTION

Data mining for education is a field with decades of history and evolved with time. This has made a lot of progress since its inception. It is always an emerging discipline, as with the history of mankind, the craze for knowledge and getting to know things is ever increasing with time. Humans are always concerned about drawing conclusions, getting to know and discover new methods to carry out some unique processes, and tackle large-scale data that result in favorable and admirable outcomes to better your educational settings and revitalize your knowledge-seeking scenarios. These will also help in conveying knowledge to students and getting them better understand, operate the way they wanted to learn and adopt. Another terminology under this context is educational data mining [1], which involves the rapid growth of an era of education that is all about making learning and understanding easy. This is all about exploring diverse ways of education, making knowledge more explorative, and mending new ways of seeking knowledge. This has implications with human nature, psychology, and inner instincts. Educational data mining has been perpetrated into different disciplines. This has a lot to do with cognitive sciences, intelligence, computer science and its interaction with humans, machine learning, machine implications within human lives, artificial intelligence, cognitive psychology, as well as, learning psychometrically, considering educational psychology, and statistics; with its methods and their implications[2].

Educational data mining demands consideration of interactive learning environments, collaborative learning, computer-oriented learning environments, as well as, administrative data available from the schools and universities that can lead to meaningful levels and responses regarding every stage.

Here, the chosen technique for data mining needs is cluster sampling technique. The concerned cluster analyzed here, is the gross percentage of students who get enrolled in primary and secondary schools, out of total school-going age children and teenagers. The target pollution involves school-going children out of the total population of entire China. Considering the Chinese population being a large estimation, so it should be narrow down to the target

population; a population that possesses a concerned characteristic that the research study carves for. Then the analysis is conducted following all the steps to perform the cluster sampling analysis.
Another terminology globally recognized is Knowledge Discovery in Databases (KDD), which involves extracting out useful information and contribution to the cause of exploration of knowledge through large datasets. According to Jiawei Han and Harvey J. Miller, the process of Knowledge Discovery consists of Data Cleaning [3], Data Integration [4], Data Selection, Transformation [5], Data Mining, and Pattern Evaluation Phases [6].
Data mining is all about extracting hidden patterns using clustering or other sampling techniques. Then this data is further employed in machine learning and predictive analysis. This analyzes data warehouses, benefiting effective decision-making. Weiss and Indurkhya have interpreted the situation as "Data Mining is a hunt for valuable information through large volumes of data", this involves digging out unknown but useful information, contributing to the evolution of knowledge, pattern finding, with the application of statistical and mathematical techniques [7].
There is a society that aims to collaborate and support educational data mining that is known as the International Educational Data Mining Society. This works for the scientific development of new ways for new and existing disciplines.

2 OBJECTIVES

- Improvise educational environment
- Work for the development of educational backgrounds and learning setups
- Strategizing educational data mining for better results and orientation

3 HISTORICAL PERSPECTIVE

The data mining term was coined in the year 1990 and it combines various disciplines including database management systems [8], Statistics, Artificial Intelligence [9], and Machine Learning [10]. The purpose behind its orientation and use is identical as it involves generating and aligning useful patterns by employing different algorithms.
Before the emergence of Statistics and Machine Learning techniques, algorithms [11] show the best results for numerical data but these are now developed for non-numerical or qualitative data as well. The core output is still the same, to provide meaningful results and evolve research methods, or knowledge creation (Dangeti, 2017).
In bygone days, data mining algorithms involve only statistical techniques, but now, with the addition of Artificial Intelligence, Machine learning techniques, and Pattern recognition [12], heterogeneous concepts and anomalies could also be detected within data warehouses.
Now, with the contribution of research methods and computational techniques, data mining applications could be employed in various fields like health care, finance, telecommunication, fraud detection, risk analysis, education, etc. Due to the complexities and anomalies arising with the advancement of technology, data mining has a more diverse and complex nature tackling different data formats, network resources, and complex databases.

4 DATA MINING TYPES

Data mining comes in several types depending upon the orientation of data. The first of such kind is Pictorial data mining, which is all about digging out pictures meaningful to your hunt. These data mining types are classified and detailed in below-pasted table, shown in table 1

5 RELATED WORKS

Cristobal Romero with his research fellows wrote a paper named "Data mining in Education" that discussed data mining in education as a research-based study and is considered it be an interdisciplinary research field. They considered it to be a way for developing methods of data formulation and advancements in educational fields. They categorized it as a way to give a better understanding to students and specifying objectives and characteristics of studying different educational perspectives [13].
A research study conducted by R. Tamilselvi and her fellow researcher with the topic of "Overview of Data Mining Techniques and Applications", has entailed all about data mining, its types, applications, and methods of data mining. They defined data mining as a science of extracting information from raw data or datasets or databases. They predicted the use of machine learning techniques, statistical analysis and procedures, and database applications.
They gave an overview of data mining, its types, and algorithms to discuss various data mining techniques. Furthermore, Decision Trees, Clustering, statistical algorithms for data mining are also discussed in their research. With the infiltration of various algorithms and clustering techniques, the use of these algorithms, their applications, merits, and demerits could also be suggested [14].
Kenneth R. Koedinger and his research fellows have studied Educational data mining (EDM) as a building block of various disciplines and educational technologies. They considered it as an answer to questions of cognition, metacognition, motivation, affect, language, social discourse, etc. This is done using intelligent tutoring systems [15], educational games, and simulations, as well as, discussion forums.
Educational Data Mining has discussed concerns of the psychology of learning with every aspect of assessment, model discovery, motivation, and metacognition on learning, as well as collaborative learning. This is evaluated with the assessment of how different statistical methods could be used to go for data mining, how cognitive models could be interpreted from data and used for improvement, how models of student effect could be used with dialog- based tutoring system, and how machine learning techniques could be applied to produce meaning outcomes that support learning at the grass-root level.
Another research study was done by Sonali Agarwal and her research fellows conducted with the topic of "Data Mining in Education, with the discussions of data classification and decision tree approach". They consider data mining as an emerging technique and consider this to be an evolutionary characteristic of knowledge building. They considered Support Vector Machines [16] as the best classifiers, giving the output of maximum accuracy and minimum root mean square error [17].
Another research study conducted by Ryan S. J. d. Baker [18] with the topic of "Data Mining for Education", has discussed Knowledge

Table 1: Different Data Mining types, Their Goals, and Applications

Data mining type	Goal	Applications
Pictorial data mining	Meaningful picture hunt	Imagery description, a better understanding
Predictive data mining	Model prediction	predict unknown or future values of interest
Descriptive Data Mining	Description of data	Useful meanings are given to raw data
Text mining	Text retrieval	Artificial Intelligence, Natural Language Processing, as well as transforming text into and organized data sets
Social media mining	Finding useful posts	Web-based platforms and Social media channels.
Audio and Video mining	Information retrieval	Audio and video sharing and information sharing

Discovery in Databases, as a way to discover useful information by the exploration of large datasets. He highlighted the application of retail sales, bioinformatics, and counter-terrorism, being the most discussable and significant applications of data mining. His research also studied educational data mining as a way to develop methods for evolution and contribution to the knowledge base.

6 DATA MINING PRE-REQUISITES

Before jumping to Data mining, some steps and problem orientation are needed. The first step involves defining the problem, what we have to do and what needs to be followed for critical observation. For the said purpose, first, there is a need to clarify and understand what objectives need to be followed, what resources could be used, and what would be your desired outputs.

Then there comes the quality mechanism. These are the steps to verify the quality of your data before employing data mining techniques to ensure there are no anomalies or missing values lying in the data so that it might come out as your desired output. If there are anomalies, you need to perform data cleaning. Data cleaning also account for the data transformation [5] that scrutinizes your data for final processing. Scrutinizing of the data involves summarizing and generalization with manipulations regrading low-level data with higher-level conceptualizations, and normalization to define assigned set of attributes.

Then model your data according following several defining conditions, using inductive reasoning techniques and algorithms. Then segment your data with the implementation of statistical clustering techniques. This will divide data into clusters and will differentiate every character of your data. Going ahead, do Link analysis to identify associations among different variables in your dataset, and detect any deviations that interrupt in the process of clustering segmentation.

7 DATA MINING METHODS

Methods used for the analysis and interpretation of results for this research study involve some typical and materialistic patterns. Here in this study, we are going to focus on the Clustering technique of data mining.

The prime focus of this study is to discuss the Clustering technique for data mining. This is used to make clusters of data or data points with the same characteristics. This might not be confused with classification. Because it allows you to put objects into predefined classes. Classification might classify objects or observations into different classes and let them work separately with their dimensions.

The clustering technique involves analyzing data points without considering any label. Clustering will generate labels for your data and separates those data points concerning their labels. Within clusters, you will find similarities, but outside of these clusters, there will be severe dissimilarities.

The aim of this study is to conduct an unsupervised classification of a very large data set. For this classification, we need to pass our data set through a classification channel that has restrictions for large data sets. Let us first reduce the complexity of the computations of the data by defining the targeted population and sampled population, as well as, the study population, possessing our needed characteristics.

At first, estimations of the simple random sample from the entire data set are performed. This random sample is derived from the entire population of China, we narrow it down to our study population of school-going children.

This accounts for several hundred million children and teenagers. These are studied with the entirety of their nature and further categorized into clusters of primary and secondary school enrollments. We performed an analysis to check their significance and handled the gross percentage of enrollments over the years to primary and secondary level schools. We handled large clusters and identified associations among these clusters.

In short, we have an aim to propose a sampling and subsampling strategy for clustering large data sets using our available resources. We used this measure and defined our population into two clusters. One of which contains the gross % of students enrolled in primary schools and the other includes the gross % of students enrolled in secondary schools.

Some other data mining techniques include classification, prediction, and sequential pattern. Their application in comparison with other methods is thoroughly entailed below, shown in table 2

The purpose here is to assess the data sets using the chi-square test of association and assess with what effect, the students get to enroll in secondary classes after passing primary classes, and with what effect, and the students get enrolled in primary classes when they reach the school-going age.

Table 2: Data Mining Techniques with Comparison and Implications

Techniques	Applicative fields ordataset	Implications
Association	Academic or non-academic dataset involving patterns, or links	Finding patterns, the association between transaction and its items
Clustering	Defined classes or undefined classes within dataset to signify characters	Clusters of data points with similarities and differences
Classification	Dataset with pre-defined classes or groups	Categorization of data points with pre-defined groups or characteristics
Prediction	Data with future motives like weather forecast, fashion, or season data forecasts	Associations and predicting future outcomes
Sequential patterns	Accounting datasets with proportionality trends	Transactional data and identification of trends
Neural networks	Complex datasets with neurological meanings or applications	Deriving meanings from complex data and can be used to detect patterns
Decision trees	Decisive datasets withsome sorts of decisions	Defining sets of decisions using CART and CHAID

8 TRANSACTIONAL DATA MINING TECHNIQUE

The first such technique is the Visualization technique that involves exploratory data analysis and collecting information about large databases. These are applied with the addition and integration of humans within this data mining process. These techniques work on large data sets and provide interactive outputs. Some of the visualization techniques include scattering plots, matrices, parallel coordinates, and geometric techniques including icon-based technique, hierarchical technique, web-based technique, graph-based technique, and dynamic techniques. These are all used for transactional data that's mostly dealt with at retail stores, industries, and warehouses.

Another application for this type of data is the Artificial intelligence technique, involving pattern recognition, machine learning, and neural networks, as a whole. There are also some other implications of AI that involve knowledge acquisition, knowledge representation, and search. These might not seem far away, but these are related to algorithms or processes of various data mining techniques.

The decisive form or opinion making mining technique is a Decision tree approach, which involves a non-linear data structure, starting from the root node and ending with the leaf node. This is all about reaching and managing the sets of decisions. Different decision tree methods include Classification and Regression Trees (CART) and Chi-Square Automatic Interaction Detection (CHAID). These are employed for the classification of data points or observations. These might differ concerning the orientation or outputs of their decision but processing will always be the same.

9 DISCUSSION

Data mining techniques with educational implications involve analyzing models using Decision Trees, Naive Bayes theorem, Linear Regression to check and assess associations, Description algorithms, and Cluster techniques. Here in this study, Secondary data is analyzed, taken through World Bank data source; a reliable and rigorous data source. This could be accessed by visiting the links mentioned here (https://data.worldbank.org/indicator/SE.PRM.ENRR?end=2020&start=1970&view=chart, https://data.worldbank.org/indicator/SE.SEC.ENRR?end=2020&start=1970&view=chart)

After extracting necessary data, linear regression analysis and chi-square analysis is performed.

Defining dataset and model: Y = a + b X, Y = Gross % of students getting enrolled in Secondary schools X = Gross % of students getting enrolled in Primary schools.

With this model, significant trend and sheer strength are visible regarding R-Square and Adjusted R-square criteria. Defining link between primary and secondary schools enrollment and link building to define the objective of the study, the results are shown in table 3 and table 4

With linear variables intact, this concludes that Linear-by-Linear association within these two clusters is significant and we may conclude that the likelihood of students getting enrolled in secondary classes has increased over the years in China. The graph pasted below shown in Fig 1 proves this claim right that how students are getting enrolled in China, over the years from 1970 to 2010. With orange bars showing frequency of students getting enrolled in secondary schools of China after passing our primary education has approached primary schools' enrollment in the recent past up till 2010.

10 RESULTS AND CONCLUSION

Talking about the results and outcomes, there is a need to develop a data-oriented and data-based instructional derivation that can contribute to the efforts of data mining with their implication. Though these results are achievable with specific research projects and educational setups still it is likely to go for these with little effort and little changes in educational setups and learning management systems. This could be more achievable and more demanding if student model parameters are used with other student populations, and results are drawn with a different context. Another approach is to promote the exchange of data and models within boundaries and within systems to achieve harmony.

With this study, the concluded remarks involves setting new trends in secondary schools enrollment as with the students passing out

Table 3: The statistcal analysis results

Regression Statistics	
Multiple R	0.577041
R Square	0.332976
Adjusted R Square	0.315873
Standard Error	12.99929
Observations	41

Table 4: Chi-Square Tests

	Value	df	Asymp. Sig. (2- sided)
Pearson Chi-Square	1122.000	1089	.238
Likelihood Ratio	239.793	1089	1.000
Linear-by-Linear Association	12.695	1	.000

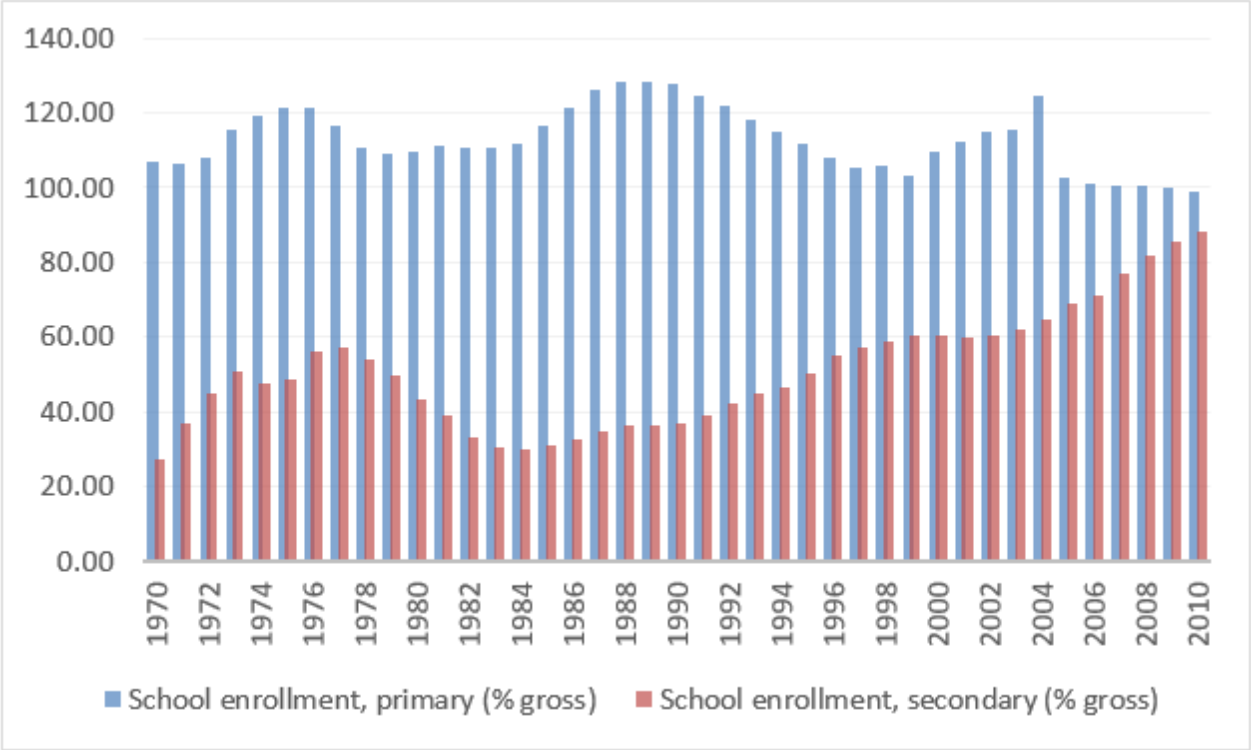


Figure 1: Gross % of students enrolled in primary and secondary schools of China

primary school education, over the years. This has improved a lot over the past in recent years. Now, with aspects of graph pasted above, tackling frequency of students getting enrolled in recent past has shown a significant increase, and brought it to almost 90% of the students enrolled in primary schools.

As with the study's discussion, educational data mining is a vast field with thorough implications for the evolution of the knowledge base and this relates to several areas of research involving internet-based learning, web data mining or finding out facts through the web, text mining, etc. It involves the use of techniques that dig out useful information from large datasets. The presented data mining with its types, techniques, measurements, and improvement scope to give what it could be to better the process of data mining, as well as getting things accumulated for those who are unaware of

data mining. Still, there is a lot of scope and space left in the field of data mining and there is much more to come in the future.
With different techniques like decision trees and visualizations, class results and performances could be measured and assessed. This classifies students based on performance. Other techniques of classification and clustering could be used to highlight below the attendance students, showing poor performance students and those who have got unsatisfactory remarks. The overall purpose is to assess the student's academic performance.

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